Data Cleaning

1. Import Libraries & Loading dataset
2. Dropping columns
3. Missing value treatment
    - Filling with values
    - Dropping of Rows
    - Imputing the Values
4. Duplicate removal
5. Outlier detection and treatment

Importing Libraries

In [1]:
```
import pandas as pd
import numpy as np
df=pd.read_csv("Salaries.csv")
df.head()
```

Out[1]:

	Unnamed: 0	rank	discipline	yrs.since.phd	yrs.service	sex	salary
0	1	Prof	B	19.0	18.0	Male	139750
1	2	Prof	B	20.0	16.0	Male	173200
2	3	AsstProf	B	4.0	3.0	Male	79750
3	4	Prof	B	45.0	39.0	Male	115000
4	5	Prof	B	40.0	41.0	Male	141500

Dropping of Unnecessary columns

In [2]:
```
df.drop('Unnamed: 0', axis=1, inplace=True)
df.head()
```

Out[2]:

	rank	discipline	yrs.since.phd	yrs.service	sex	salary
0	Prof	B	19.0	18.0	Male	139750
1	Prof	B	20.0	16.0	Male	173200
2	AsstProf	B	4.0	3.0	Male	79750
3	Prof	B	45.0	39.0	Male	115000
4	Prof	B	40.0	41.0	Male	141500

Missing Value Treatment

Checking for Null Values

In [3]:
```
df.isnull().sum()
```

Out[3]:
```
rank              2
discipline        0
yrs.since.phd     1
yrs.service       1
sex               1
salary            0
dtype: int64
```

In [4]:
```
# Separating the numerical and categorical columns
numeric_cols = df.select_dtypes(include=np.number).columns
categorical_cols = df.select_dtypes(exclude=np.number).columns
print(numeric_cols)
print(categorical_cols)
```

```
Index(['yrs.since.phd', 'yrs.service', 'salary'], dtype='object')
Index(['rank', 'discipline', 'sex'], dtype='object')
```

In [5]:
```
df.dtypes
```

Out[5]:
```
rank               object
discipline         object
yrs.since.phd     float64
yrs.service       float64
sex                object
salary              int64
dtype: object
```

Filling Numeric columns with median

In [6]:
```
for col in numeric_cols:
    if df[col].isnull().sum() > 0:
        df[col] = df[col].fillna(df[col].median())
```

Filling categorical columns with mode

In [7]:
```
for col in categorical_cols:
    if df[col].isnull().sum() > 0:
        df[col]=df[col].fillna(df[col].mode()[0])
```

In [8]:
```
df.isnull().sum()
```

Out[8]:
```
rank             0
discipline       0
yrs.since.phd    0
yrs.service      0
sex              0
salary           0
dtype: int64
```

Dropping of Rows

In [9]:
```
df.dropna(inplace=True)
```

Filling with Simple Imputer

In [10]:
```
from sklearn.impute import SimpleImputer
mean_imputer = SimpleImputer(strategy="mean")
df[numeric_cols] = mean_imputer.fit_transform(df[numeric_cols])
```

In [11]:
```
cat_imputer = SimpleImputer(strategy="most_frequent")
df[categorical_cols] = cat_imputer.fit_transform(df[categorical_cols])
```

In [12]:
```
df.isnull().sum()
```

Out[12]:
```
rank             0
discipline       0
yrs.since.phd    0
yrs.service      0
sex              0
salary           0
dtype: int64
```

Duplicate Removal

In [13]:
```
duplicate_count = df.duplicated().sum()
print(duplicate_count)
```

```
7
```

In [14]:
```
df = df.drop_duplicates()
```

In [15]:
```
df.duplicated().sum()
```

Out[15]:
```
np.int64(0)
```

Outlier Detection and Treatment

In [16]:
```
for col in numeric_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
```

```
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]
outliers
```

Out[16]:

	rank	discipline	yrs.since.phd	yrs.service	sex	salary
44	Prof	B	38.0	38.0	Male	231545.0
252	Prof	A	29.0	7.0	Male	204000.0
367	Prof	A	43.0	43.0	Male	205500.0

In [17]:
```
outliers.index
```

Out[17]:
```
Index([44, 252, 367], dtype='int64')
```

In [18]:
```
df.drop(outliers.index,axis=0,inplace=True)
```

Saving the cleaned data

In [19]:
```
df.to_csv("salaries_cleaned.csv",index=False)
```

Done!!!